Assessment 3: RNNs for stock price prediction

Start Assignment

- Due Saturday by 22:59
- Points 100
- Submitting a file upload

Weighting

This assessment is worth **25%** of your overall grade.

Course Learning Outcomes

- CLO 1: Adapt and apply techniques to improve Neural Networks.
- CLO 2: Apply strategies to design and implement a machine learning project using Deep Learning techniques.
- CLO 3: Interpret future developments in Deep Learning based on core concepts presented.
- CLO 4: Analyse Neural Network architectures including Convolutional Neural Networks.
- CLO 5: Critically review and integrate Neural Network solutions to address real-life problems.

Task Description

Word count/length: 6-8 pages including text, references and diagrams.

Check the **Assignment 3 Requirements. (https://myuni.adelaide.edu.au/courses/94658/files/16214701/download)**

Purpose:

- Design a project using Recurrent Neural Networks, fundamental to predicting sequential events
- Compare performance of RNN architectures for stock price prediction, based on real historical data
- Use knowledge of RNN structure to design experimental test cases
- Report and interpret experimental results in a clear written format

Assessment Criteria:

Assessment is based on a summary report, comprising:

- Introduction and problem background, including comparison and selection of competing approaches
- Description of the method you have developed and key design choices made
- Experimental analysis and testing of your method
- Evidence of correct implementation of method
- Reflection on experimental results and future improvements

Academic Integrity

It is your responsibility to ensure that any work you submit is your own.
You can use the **Turnitin Originality Checker (https://myuni.adelaide.edu.au/courses/24800/pages/turnitin-students/)** before you submit your work.

Rubric: Assessment 3 RNNs for stock price prediction

Criteria	Ratings					Pts
Introduction and Background	**15 to >12.75 Pts High Distinction** Introduction clearly summarises the problem and what you have done. Background includes accurate comparison with leading competitors.	**12.75 to >11.25 Pts Distinction** Introduction clearly summarises the problem and what you have done. Background includes comparison with some competitors.	**11.25 to >9.75 Pts Credit** Introduction clearly states the problem and what you have done. Background has flawed comparison with competitors.	**9.75 to >7.5 Pts Pass** Introduction states the problem but with some errors. Background includes competing methods but lacks comparison.	**7.5 to >0 Pts Fail** Introduction does not clearly state the problem or what was done. Background is missing, erroneous or irrelevant.	15 pts
Method description	**20 to >17.0 Pts High Distinction** Description includes all key points of the method and would be reproducible.	**17 to >15.0 Pts Distinction** Description covers the method but some steps are unclear.	**15 to >13.0 Pts Credit** Description covers most of the method but misses some key steps.	**13 to >10.0 Pts Pass** Description covers some of the method but some is missing or unclear.	**10 to >0 Pts Fail** Description is missing or not clear.	20 pts
Method implementation	**30 to >25.5 Pts High Distinction** Implementation works correctly and is easy to read.	**25.5 to >22.5 Pts Distinction** Implementation works correctly.	**22.5 to >19.5 Pts Credit** Implementation works correctly in most cases.	**19.5 to >15.0 Pts Pass** Implementation functions but has significant flaws.	**15 to >0 Pts Fail** Implementation is incorrect or incomplete.	30 pts

Criteria	Ratings					Pts
Experiments and analysis	**5 to >4.25 Pts High Distinction** Clear description of experiments and the aims of your experimental design. Results displayed in a compact and easy to read format. Analysis of results draws accurate conclusions about the method.	**4.25 to >3.75 Pts Distinction** Experimental results included and clearly explained. Some analysis of significance of results.	**3.75 to >3.25 Pts Credit** Experimental results included and completely explained.	**3.25 to >2.5 Pts Pass** Experimental results included but missing explanation or definition.	**2.5 to >0 Pts Fail** Experiments are missing or irrelevant. No results or relevant analysis provided.	5 pts
Reflection on project	**30 to >25.5 Pts High Distinction** Clear summary of major project design choices. Ideas for future work based on experimental evidence.	**25.5 to >22.5 Pts Distinction** Clear summary of major project design choices. Ideas for future work included.	**22.5 to >19.5 Pts Credit** Project is summarised along with ideas for future work.	**19.5 to >15.0 Pts Pass** Evidence of lessons learnt from project.	**15 to >0 Pts Fail** No evidence of engaging with the project..	30 pts
						Total points: 100